

Balance a One-Step Equation



Grade 6 • Mathematics
NC.6.EE.7

Write and solve one-step equations.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

Solve $9x = 81$. Show the same operation on both sides, then substitute to check.

Divide both sides by 9: $x = 9$. Check: $9 \times 9 = 81$.



Build a second model together

Solve $x + 11/2 = 35/2$. Show the same operation on both sides, then substitute to check.

Discuss: What should the first step find?



Your turn • 1

Solve $8x = 60$. Show the same operation on both sides, then substitute to check.

Represent → solve → explain → check



Your turn • 2

Solve $x + 4 = 25/2$. Show the same operation on both sides, then substitute to check.

Represent → solve → explain → check



Your turn • 3

Solve $9x = 108$. Show the same operation on both sides, then substitute to check.

Represent → solve → explain → check



Your turn • 4

Solve $x + 5 = 29/2$. Show the same operation on both sides, then substitute to check.

Represent → solve → explain → check



Exit question • Explain independently

Solve $2x = 4$. Show the same operation on both sides, then substitute to check.

Include a check. What would you teach someone else?



Teacher review and next steps

1. $x = 15/2$

2. $x = 17/2$

3. $x = 12$

4. $x = 19/2$

Exit: $x = 2$

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

